Analysis of the Min-Sum Algorithm for Packing and Covering Problems via Linear Programming

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***Abstract*— Message-passing algorithms based on belief-propagation (BP) are successfully used in many applications, including decoding error correcting codes and solving constraint satisfaction and inference problems. The BP-based algorithms operate over graph representations, called factor graphs, that are used to model the input. Although in many cases, the BP-based algorithms exhibit impressive empirical results, not much has been proved when the factor graphs have cycles. This paper deals with packing and covering integer programs in which the constraint matrix is zero-one, the constraint vector is integral, and the variables are subject to box constraints. We study the performance of the min-sum algorithm when applied to the corresponding factor graph models of packing and covering linear programmings (LPs). We compare the solutions computed by the min-sum algorithm for packing and covering problems to the optimal solutions of the corresponding LP relaxations. In particular, we prove that if the LP has an optimal fractional solution, then for each fractional component, the min-sum algorithm either computes multiple solutions or the solution oscillates below and above the fraction. This implies that the min-sum algorithm computes the optimal integral solution only if the LP has a unique optimal solution that is integral. The converse is not true in general. For a special case of packing and covering problems, we prove that if the LP has a unique optimal solution that is integral and on the boundary of the box constraints, then the min-sum algorithm computes the optimal solution in pseudopolynomial time. Our results unify and extend recent results for the maximum weight matching problem and for the maximum weight independent set problem.**

***Index Terms*— Belief propagation (BP), message-passing algorithms, min-sum algorithm, max-product algorithm, dynamic programming, linear programming (LP), combinatorial optimization, packing problems, covering problems, factor graphs, graph cover.**

I. INTRODUCTION

WE CONSIDER optimization problems over the integers called *packing* and *covering* problems. Many optimization problems can be formulated as packing problems including maximum weight matchings and maximum weight independent sets. Optimization problems such as minimum weight set-cover and minimum weight dominating set are special cases of covering problems. The input for both types of problems consists of an $m \times n$ zero-one constraint matrix $\mathbf{A}$, an integral constraint vector $\mathbf{b}$, an upper bound vector $\mathbf{X} \in \mathbb{N}^n$, and a weight vector $\mathbf{w} \in \mathbb{R}^n$. An integral vector $\mathbf{x} \in \mathbb{N}^n$ is an *integral packing* if $\mathbf{0} \leq \mathbf{x} \leq \mathbf{X}$ and $\mathbf{A} \cdot \mathbf{x} \leq \mathbf{b}$. In a packing problem the goal is to find an integral packing that minimizes $\mathbf{w}^T \cdot \mathbf{x}$. An integral vector $\mathbf{x} \in \mathbb{N}^n$ is an *integral covering* if $\mathbf{0} \leq \mathbf{x} \leq \mathbf{X}$ and $\mathbf{A} \cdot \mathbf{x} \geq \mathbf{b}$. In a covering problem the goal is to find an integral packing that maximizes $\mathbf{w}^T \cdot \mathbf{x}$.

Packing and covering problems generalize problems that are solvable in polynomial time (e.g., maximum matching) and problems that are NP-hard and even NP-hard to approximate (e.g., maximum independent set). The hardness of special cases of these problems imply that general algorithms for packing/covering problems are heuristic in nature. Two heuristics that are used in practice to solve such problems are linear programming (LP) and belief-propagation (BP).

Linear programming deals with optimizing a linear function over a convex polyhedral subset of Euclidean space (i.e., a subset defined by the intersection of a finite number of half-spaces) [1]. Perhaps the simplest way to utilize linear programming in this setting is to solve the LP relaxation of the integer problem (i.e., relax the restriction that $\mathbf{x} \in \mathbb{N}^n$ to the restriction $\mathbf{x} \in \mathbb{R}^n$). If the result happens to be integral, then we are lucky and we have found an optimal integral packing or covering. A great deal of literature deals with characterizing problems for which this method works well (e.g., works on total unimodularity [2]). In fact, LP decoding of error correcting codes works in the same fashion and has been proven to work well in average [3], [4].

Belief-propagation[1] is an algorithmic paradigm that deals with inference over graphical models [5]. The graphical model that corresponds to packing/covering problems is a bipartite graph that represents the zero-one matrix $\mathbf{A}$. We focus on a common variant of belief-propagation that is called the min-sum algorithm (or the max-product algorithm). In the variant we consider, the initial messages are all zeros and messages are not attenuated.

Our main result is a proof that the min-sum algorithm is not better than the heuristic based on linear programming. This proof holds for every instance of packing and covering problems described above with respect to the min-sum algorithm with zero initialization and no attenuation.

A. Previous Work

BP-type message-passing algorithms have been invented multiple times (see [5]–[7]) and have many

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[1] "Belief propagation" (BP) is the name given by Pearl [5] for a message-passing algorithm that computes exact probabilities on tree models. We generally refer by *BP-based* or *BP-type* algorithms to message-passing algorithms in a setting of BP (even when applied on cyclic graphs).

applications (see [8]). In this paper we focus on a common variant of BP called the *min-sum algorithm*. In his thesis, Wiberg [9] gave the first clear description of the min-sum algorithm as a generalization of Viterbi decoding [7]. A BP-based algorithm is a message-passing algorithm in which messages are sent along edges of a graph called the *factor graph*. The factor graph of packing/covering problems is a bipartite graph that represents that constraint matrix $\mathbf{A}$. In essence, value computed by the min-sum algorithm for x_i equals the outcome of a dynamic programming algorithm over a path-prefix tree rooted at the vertex corresponding to the ith column of $\mathbf{A}$. Since dynamic programming computes an optimal solution over trees, the min-sum algorithm is optimal when the factor graph is a tree [5], [9]. A major open problem is to analyze the performance of BP (or even the min-sum algorithm) when the factor graph is not a tree. Execution of algorithms based on BP over graphs that contain cycles is often referred to as *loopy BP*.

Vontobel and Koetter [10] were the first to notice the close connection between BP-type algorithms to LP relaxations. Recently, a few papers have studied the usefulness of the min-sum algorithm for solving optimization problems and compared its solutions to solutions obtained by linear programming. Such a comparison for the maximum weight matching problem appears in [11]–[13] with respect to constraints of the form $\sum_{u \text{ neighbor of } v} x_{(u,v)} \leq 1$ for every vertex v. Loosely speaking, the main result that they show for maximum weighted matching is that the min-sum algorithm is successful if and only if the LP heuristic is successful. The sufficient condition states that if the LP relaxation has a unique optimal solution and that solution is integral, then the min-sum algorithm computes this solution in pseudo-polynomial time. The necessary condition states that if the LP relaxation has a fractional optimal solution, then the min-sum algorithm fails. In [14], the min-sum algorithm for the maximum weighted independent set problem was studied with respect to constraints $x_u + x_v \leq 1$ for every edge (u, v). They prove an analogous necessary condition and present a counter-example that proves that the sufficient condition (i.e., unique optimal solution for the LP that is integral) does not imply the success of the min-sum algorithm. The performance of the min-sum algorithm has been also studied for computing shortest s-t paths [15] and min-cost flows [16], [17].

B. Our Results

The results in this paper extend and generalize previous necessary conditions for the success of the min-sum algorithm in the case where messages are initialized to zero. This necessary condition implies that, compared to the LP heuristic, the min-sum algorithm is not a better heuristic for solving packing and covering problems. Our contributions can be summarized as follows:

(i) We consider a unified framework of packing and covering problems. Previous works deal with a zero-one constraint matrix $\mathbf{A}$ that has two nonzero entries in each column (for maximum weight matching [11]–[13]) or two nonzero entries in each row (for maximum weight independent set [14]). Our results hold with respect to any zero-one constraint matrix $\mathbf{A}$.

(ii) We allow box constraints, namely $x_i \in \{0, 1, \ldots, X_i\}$. Previous results consider only zero-one variables.

(iii) Our oscillation results hold also when the LP relaxation has multiple solutions. To obtain such a result, we consider the set of optimal values computed by the min-sum algorithm at each variable (rather than declare failure if there are multiple optimal values). We compare these sets with the optimal solutions of the LP relaxation and show a weak oscillation between even and odd iterations.

(iv) The analogous result for covering LPs is obtained by a simple reduction (see Claim 3) that applies complementation; this generalizes reductions from maximum matchings to minimum edge covers [13].

(v) We present a unified proof based on graph covers. This method also enables us to prove convergence of the min-sum algorithm under certain restrictions (see Theorem 19 in Appendix).

Our problem setting captures (as special cases) the problems of maximum weight (perfect and non-perfect) b-matchings, maximum weight independent set, and minimum weight r-edge cover dealt separately in previous works [11]–[14]. Further problem instances that fall within our problem setting include, for example, the maximum weight set packing and its variants. Variants of the maximum weighted set packing problem are used to model, for example, the winner determination problem in combinatorial auctions (see [18]).

C. Techniques

The main challenge in comparing between the min-sum algorithm and linear programming is in finding a common structure that captures both algorithms. It turns out that graph covers are a common structure. Graph covers have been used previously to analyze iterative message passing algorithms [10], [19]. In the context of optimization problems, 2-covers have been used in [12] to reduce matchings in general graphs to matchings in bipartite graphs [11]. Bayati *et al.* [12] write that their "proof gives a better understanding of the often-noted but poorly understood connection between BP and LP through graph covers." We further clarify this connection by using higher order covers that capture fractional optimal LP solutions, as suggested by Ruozzi and Tatikonda [20]. Graph covers not only capture LP solutions but also solutions computed by the min-sum algorithm. In fact, the min-sum algorithm performs the same computation over any graph cover because it operates over a path-prefix tree of the factor graph. Hence we make the mental experiment in which the min-sum algorithm is executed over a graph cover in which all the basic feasible solutions are integral. We avoid the problems associated with loopy-BP by considering a graph cover, the girth of which is much larger than the number of iterations of the min-sum algorithm. Thus the execution of the min-sum algorithm is equivalent to a dynamic programming over subtrees induced by balls in the graph cover. This mental game justifies a dynamic programming interpretation of the outcome of the min-sum algorithm. The dynamic programming algorithm makes a "local" decision based on balls, the radius of which is twice the number of iterations. The LP solution, on the other hand, is a global solution.

The proof proceeds by creating "hybrid" solutions (by applying "pivots") that either refute the (global) optimality of the LP solution or the (local) optimality of the dynamic programming solution over the ball in the graph cover.

In contrast, the techniques in the previous works (e.g., in the cases of maximum weight matching and maximum weight independent set [11]–[14]) are based on perturbation arguments. Perturbations are applied to the optimal solution of the LP, and are fitted for each of the problems. Our pivot arguments can be interpreted as generalizations of the perturbation techniques.

II. Preliminaries

A. Graph Terminology and Algebraic Notation

1) Algebraic Notation: Let $\mathbb{N}$, $\mathbb{N}_{>0}$, $\mathbb{R}$, and $\mathbb{R}_{\geq 0}$ denote the set of natural numbers (including 0), the set of positive integers, the field of real numbers, and the set of non-negative real numbers, respectively. We denote vectors in bold, e.g., $\mathbf{x}, \mathbf{z}$. We denote the ith coordinate of $\mathbf{x}$ by x_i, e.g., $\mathbf{x} = (x_1, \ldots, x_n)$.

For a vector $\mathbf{x} \in \mathbb{R}^n$, let $\|\mathbf{x}\|_1 \triangleq \sum_i |x_i|$ denote the ℓ_1 norm of $\mathbf{x}$. For two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$, let $\mathbf{x} \cdot \mathbf{y} \triangleq \sum_i x_i \cdot y_i$ denote the standard inner product of the two vectors. The cardinality of a set S is denoted by $|S|$. We denote by $[n]$ the set $\{0, 1, 2, \ldots, n\}$ for $n \in \mathbb{N}$. For a set $S \subseteq [n]$, we denote the projection of the vector $\mathbf{x}$ onto indices in S by $\mathbf{x}_S \in \mathbb{R}^{|S|}$.

A vector is *rational* if all its components are rational. Similarly, a vector is *integral* if all its components are integers. A vector is *fractional* if it is not integral, i.e., at least one of its components is not an integer.

Let $\mathbf{X} \in \mathbb{N}^n$ denote a non-negative integral vector. Denote the Cartesian product $[X_1] \times \cdots \times [X_n]$ by $\mathrm{ZBox}(\mathbf{X})$. Similarly, denote the Cartesian product $[0, X_1] \times \cdots [0, X_n]$ by $\mathrm{RBox}(\mathbf{X})$. Note that vectors in $\mathrm{ZBox}(\mathbf{X})$ are integral.

2) Graph Terminology: Let $G = (V, E)$ denote an undirected simple graph. Let $\mathcal{N}_G(v)$ denote the set of neighbors of vertex $v \in V$ (not including v itself). Let $\deg_G(v)$ denote the edge degree of vertex v in a graph G, i.e., $\deg_G(v) \triangleq |\mathcal{N}_G(v)|$. For a set $S \subseteq V$ let $\mathcal{N}_G(S) \triangleq \bigcup_{v \in S} \mathcal{N}_G(v)$. A *path* in G is a sequence of vertices such that there exists an edge between every two consecutive vertices in the sequence. A *backtrack* in a path is a subpath that is a loop consisting of two edges traversed in opposite directions, i.e., a subsequence (u, v, u). All paths considered in this paper do not include backtracks. The *length* of a path is the number of edges in the path. We denote the length of a path p by $|p|$. Let $d_G(r, v)$ denote the distance (i.e., length of a shortest path) between vertex r and v in G, and let $\mathrm{girth}(G)$ denote the length of the shortest cycle in G. Let $B_G(v, t)$ denote the set of vertices in G with distance at most t from v, i.e., $B_G(v, t) \triangleq \{u \in V \mid d_G(v, u) \leq t\}$.

The *subgraph of G induced by $S \subseteq V$* consists of S and all edges in E, both endpoints of which are contained in S. Let G_S denote the subgraph of G induced by S. A subset $S \subseteq V$ is an *independent set* if there are no edges in the induced subgraph G_S. A graph $G = (V, E)$ is *bipartite* if V is the union of two disjoint nonempty independent sets.

B. Covering and Packing Linear Programs

We consider two types of linear programs called covering and packing problems. In both cases the matrices are zero-one matrices and the constraint vectors are positive.

In the sequel we refer to the constraints $\mathbf{x} \in \mathrm{ZBox}(\mathbf{X})$ and $\mathbf{x} \in \mathrm{RBox}(\mathbf{X})$ as *box constraints*.

Definition 1: Let $\mathbf{A} \in \{0, 1\}^{m \times n}$ denote a zero-one matrix with m rows and n columns. Let $\mathbf{x} \in \mathbb{R}^n$ denote a vector of decision variables, let $\mathbf{b} \in \mathbb{R}^m_{\geq 0}$ denote a constraint vector, let $\mathbf{w} \in \mathbb{R}^n$ denote a weight vector, and let $\mathbf{X} \in \mathbb{N}^n$ denote a domain boundary vector.

[PIP] *The integer program*

$$\max\left\{\mathbf{w}^T \cdot \mathbf{x} \;\middle|\; \mathbf{A} \cdot \mathbf{x} \leq \mathbf{b},\ \mathbf{x} \in \mathrm{ZBox}(\mathbf{X})\right\}$$

is called a packing IP*, and denoted by* PIP.

[CIP] *The integer program*

$$\min\left\{\mathbf{w}^T \cdot \mathbf{x} \;\middle|\; \mathbf{A} \cdot \mathbf{x} \geq \mathbf{b},\ \mathbf{x} \in \mathrm{ZBox}(\mathbf{X})\right\}$$

is called a covering IP*, and denoted by* CIP.

[PLP] *The linear program*

$$\max\left\{\mathbf{w}^T \cdot \mathbf{x} \;\middle|\; \mathbf{A} \cdot \mathbf{x} \leq \mathbf{b},\ \mathbf{x} \in \mathrm{RBox}(\mathbf{X})\right\}$$

is called a packing LP*, and denoted by* PLP.

[CLP] *The linear program*

$$\min\left\{\mathbf{w}^T \cdot \mathbf{x} \;\middle|\; \mathbf{A} \cdot \mathbf{x} \geq \mathbf{b},\ \mathbf{x} \in \mathrm{RBox}(\mathbf{X})\right\}$$

is called a covering LP*, and denoted by* CLP.

Let $\Pi(\mathbf{x})$ denote an integer or linear program with respect to a vector of decision variables $\mathbf{x}$. Denote by $\mathrm{OPT}\big(\Pi(\mathbf{x})\big)$ the set of optimal feasible solutions of $\Pi(\mathbf{x})$, i.e., the set of feasible decision variable vectors $\mathbf{x}$ that optimize $\Pi(\mathbf{x})$.

C. Factor Graph Representation of Packing and Covering LPs

The belief-propagation algorithm and its variant called the min-sum algorithm deal with graphical models known as *factor graphs* (see [21]). In this section we review the definition of factor graphs that are used to model covering and packing problems.

Definition 2 (Factor Graph Model of Packing Problems): A quadruple $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ is the factor graph model *of a* PIP *if:*

- *$G = (\mathcal{V} \cup \mathcal{C}, E)$ is a bipartite graph that represents the zero-one matrix $\mathbf{A}$. The set of* variable vertices *$\mathcal{V} = \{v_1, \ldots, v_n\}$ corresponds to the columns of $\mathbf{A}$, and the set of* constraint vertices *$\mathcal{C} = \{C_1, \ldots, C_m\}$ corresponds to the rows of $\mathbf{A}$. The edge set is defined by $E \triangleq \{(v_i, C_j) \mid \mathbf{A}_{ji} = 1\}$.*
- *The vector $\mathbf{X} \in \mathbb{N}^n$ defines the alphabets that are associated with the variable vertices. The alphabet associated with v_i equals $\{0, \ldots, X_i\}$.*

- *For each constraint vertex C_j, we define a packing factor function $\psi_{C_j} : \mathrm{ZBox}(\mathbf{X}) \to \{0, -\infty\}$, defined by*

$$\psi_{C_j}(\mathbf{y}) \triangleq \begin{cases} 0 & \text{if } \sum_{v_i \in \mathcal{N}_G(C_j)} y_i \le b_j \\ -\infty & \text{otherwise} \end{cases} \quad (1)$$

We denote the set of factor functions $\{\psi_{C_j}\}_j$ by Ψ.

- *For each variable vertex v_i, we define a variable function $\phi_{v_i} : [0, X_i] \to \mathbb{R}$ defined by $\phi_{v_i}(\beta) \triangleq w_i \cdot \beta$. We denote the set of variable functions $\{\phi_{v_i}\}_i$ by Φ.*

We note that (i) One can also define a factor graph for PLP; the only difference is that the alphabet associated with variable vertex v_i is the real interval $[0, X_i]$, and the range of each factor function is $\mathrm{RBox}(\mathbf{X})$. (ii) The factor functions are local in the sense that each constraint vertex C_j can evaluate the value of ψ_{C_j} based on the values of its neighbors.

A vector $\mathbf{x} \in \mathbb{R}^n$ is viewed as an *assignment* to variable vertices in $\mathcal{V}$ where x_i is assigned to vertex v_i. To avoid composite indices, we use x_v to denote the value assigned to v by the assignment $\mathbf{x}$. An integral assignment $\mathbf{x}$ is *valid* if it satisfies all the constraints, namely, $x_i \in [X_i]$ for every i and $\mathbf{A} \cdot \mathbf{x} \le \mathbf{b}$.

The factor graph model allows for the following equivalent formulation of the packing integer program:

$$\max\Big\{\sum_{v\in\mathcal{V}} \phi_v(x_v) + \sum_{C\in\mathcal{C}} \psi_C(\mathbf{x}) \;\Big|\; \mathbf{x} \in \mathrm{ZBox}(\mathbf{X})\Big\}. \quad (2)$$

If there exists at least one valid assignment, then this formulation is equivalent to the formulation:

$$\max\Big\{\sum_{v\in\mathcal{V}} \phi_v(x_v) \;\Big|\; \mathbf{x} \text{ is a valid integral assignment}\Big\}. \quad (3)$$

We may define a factor graph model for covering problems in the same manner. The only difference is in the definition of the covering factor functions, namely,

$$\psi_C(y) \triangleq \begin{cases} 0 & \text{if } \sum_{v \in \mathcal{N}_G(C)} y_v \ge b_C \\ \infty & \text{otherwise} \end{cases} \quad (4)$$

Using this factor graph model, we can reformulate the covering integer program CIP by

$$\min\Big\{\sum_{v\in\mathcal{V}} \phi_v(x_v) + \sum_{C\in\mathcal{C}} \psi_C(\mathbf{x}) \;\Big|\; \mathbf{x} \in \mathrm{ZBox}(\mathbf{X})\Big\}. \quad (5)$$

One could define a factor graph model for general LPs as well. Suppose the goal is to maximize the objective function. Then, for each constraint, the range of the factor function is $\{-\infty, 0\}$. If the constraint C is satisfied, then the value of ψ_C is 0; otherwise it is $-\infty$.

III. Min-Sum Algorithms for Packing and Covering Integer Programs

In this section we present the min-sum algorithm for solving packing and covering integer programs with zero-one constraint matrices. Strictly speaking, the algorithm for PIP is a max-sum algorithm, however we refer to these algorithms in general as min-sum algorithms. All the results in this section apply to any other equivalent algorithmic representation (e.g., max-product-type formulations). We first define the min-sum algorithms for PIPs and CIPs, and then state our main results.

A. The Min-Sum Algorithm

The min-sum algorithm for the packing integer program (PIP) is listed as Algorithm 1. The input to algorithm MIN-SUM-PACKING consists of a factor graph model $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ of a PIP instance and a number of iterations $t \in \mathbb{N}_{>0}$. Each iteration consists of two parts. In the first part, each variable vertex performs a local computation and sends messages to all its neighboring constraint vertices. In the second part, each constraint vertex performs a local computation and sends messages to all its neighboring variable vertices. Hence, in each iteration, two messages are sent along each edge (in opposite directions).

Let $\mu^{(t')}_{v\to C}(\beta)$ denote the message sent from a variable vertex $v \in \mathcal{V}$ to an adjacent constraint vertex $C \in \mathcal{C}$ in iteration t' under the assumption that vertex v is assigned the value $\beta \in \{0, \ldots, X_v\}$. Similarly, let $\mu^{(t')}_{C\to v}(\beta)$ denote the message sent from $C \in \mathcal{C}$ to $v \in \mathcal{V}$ in iteration t' assuming that vertex v is assigned the value $\beta \in \{0, \ldots, X_v\}$. Denote by $\mu_v(\beta)$ the final value computed by variable vertex $v \in \mathcal{V}$ for assignment of $\beta \in \{0, \ldots, X_v\}$.

The initial messages (considered as the zeroth iteration) have the value zero and are sent along all the edges from the constraint vertices to the variable vertices. We refer to these initial messages as the *zero initialization* of the min-sum algorithm.

The algorithm proceeds with t iterations. In Line 2a the message to be sent from v to C is computed by adding the previous incoming messages to v (not including the message from C) and adding to it $\phi_v(\beta)$. In Line 2b the message to be sent from C back to v is computed. The constraint vertex C considers all the possible assignments $\mathbf{z}$ to its neighbors in which $z_v = \beta$. In fact, only assignments that satisfy the constraint of C and the box constraints of the neighbors of C are considered. The message from C to v equals the maximum sum of the previous incoming messages (not including the message from v) among these assignments.

Finally, in Line 3 each vertex v decides locally on its outcome $\hat{x}_v$. The maximum value $\mu_v^{\max}$ of the final variable values is computed, $\mu_v^{\max} \triangleq \max\{\mu_v(\beta) \mid \beta \in [X_v]\}$. Then, the set $\delta_{v,t}$ of variable assignments that achieve the maximum value $\mu_v^{\max}$ is computed, $\delta_{v,t} \triangleq \{\beta \mid \mu_v(\beta) = \mu_v^{\max}\}$. The decision $\hat{x}_v$ of vertex v equals the minimum or maximum value in $\delta_{v,t}$ depending on the parity of t. Here, we deviate from previous descriptions of the min-sum algorithm that declare failure if $\delta_{v,t}$ contains more than one element.[2]

Algorithm MIN-SUM-COVERING listed as Algorithm 2 is based on the following reduction of the covering LP to a packing LP as follows. It is easy to write a direct min-sum formulation of algorithm MIN-SUM-COVERING.

Claim 3: Let $\mathbf{d} \triangleq \mathbf{A} \cdot \mathbf{X} - \mathbf{b} \in \mathbb{R}^m$, then

(i) *for every optimal solution $\mathbf{x}^*$ of the* PLP

$$\max\big\{\mathbf{w}^T \cdot \mathbf{x} \;\big|\; \mathbf{A} \cdot \mathbf{x} \le \mathbf{b},\; \mathbf{x} \in \mathrm{RBox}(\mathbf{X})\big\},$$

the vector $\mathbf{z}^* \triangleq \mathbf{X} - \mathbf{x}^*$ is an optimal solution of the

[2] Note that the decision on maximum (minimum) value in $\delta_{v,t}$ in even (odd) iterations is needed when $\delta_{v,t}$ contains more than one element to obtain the "weak oscillation" proved in Theorem 4.

Algorithm 1 MIN-SUM-PACKING($\langle G, \Psi, \Phi, \mathbf{X}\rangle, t$) - a Min-Sum Algorithm for a PIP $\max\{\mathbf{w}^T \cdot \mathbf{x} \mid \mathbf{A}\cdot\mathbf{x} \le \mathbf{b}, \mathbf{x} \in \mathrm{ZBox}(\mathbf{X})\}$. Given the Factor Graph Model $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ of the PIP and the Number of Iterations $t \in \mathbb{N}_{>0}$, Outputs a Vector $\hat{\mathbf{x}} \in \mathrm{ZBox}(\mathbf{X})$

1) **Initialize: for each** $(v, C) \in E$ and $\beta \in \{0, \ldots, X_v\}$ **do**
$$\mu^{(0)}_{C\to v}(\beta) \leftarrow 0$$
2) **Iterate: for** $t' = 1$ to t **do**
   a) **for each** $(v, C) \in E$ and $\beta \in \{0, \ldots, X_v\}$ **do** {variable-to-constraint message}
$$\mu^{(t')}_{v\to C}(\beta) \leftarrow \phi_v(\beta) + \sum_{C'\in\mathcal{N}(v)\setminus\{C\}} \mu^{(t'-1)}_{C'\to v}(\beta)$$
   b) **for each** $(v, C) \in E$ and $\beta \in \{0, \ldots, X_v\}$ **do** {constraint-to-variable message}
$$\mu^{(t')}_{C\to v}(\beta) \leftarrow \max\left\{ \sum_{u\in\mathcal{N}(C)\setminus\{v\}} \mu^{(t')}_{u\to C}(z_u) \;\middle|\; \begin{array}{l} \mathbf{z}\in\mathrm{ZBox}(\mathbf{X}) \\ z_v = \beta \\ \psi_C(\mathbf{z}) = 0 \end{array} \right\}$$
3) **Decide: for each** $v \in \mathcal{V}$ **do**
   a) **for each** $\beta \in \{0, \ldots, X_v\}$ **do**
$$\mu_v(\beta) \leftarrow \sum_{C\in\mathcal{N}(v)} \mu^{(t)}_{C\to v}(\beta)$$
   b) $\mu^{\max}_v \leftarrow \max\{\mu_v(\beta) \mid \beta \in [X_v]\}$
   c) $\delta_{v,t} \triangleq \{\beta \mid \mu_v(\beta) = \mu^{\max}_v\}$.
   d)
$$\hat{x}_v \leftarrow \begin{cases} \max\{\beta \mid \beta\in\delta_{v,t}\} & \text{if } t \text{ is even} \\ \min\{\beta \mid \beta\in\delta_{v,t}\} & \text{if } t \text{ is odd}\end{cases}$$

return $\hat{\mathbf{x}}$

CLP
$$\min\left\{\mathbf{w}^T\cdot\mathbf{z} \;\middle|\; \mathbf{A}\cdot\mathbf{z} \ge \mathbf{d},\ \mathbf{z}\in\mathrm{RBox}(\mathbf{X})\right\}.$$
(ii) *for every optimal solution* $\mathbf{z}^*$ *of the* CLP
$$\min\left\{\mathbf{w}^T\cdot\mathbf{z} \;\middle|\; \mathbf{A}\cdot\mathbf{z} \ge \mathbf{d},\ \mathbf{z}\in\mathrm{RBox}(\mathbf{X})\right\},$$
the vector $\mathbf{x}^* \triangleq \mathbf{X} - \mathbf{z}^*$ *is an optimal solution of the* PLP
$$\max\left\{\mathbf{w}^T\cdot\mathbf{x} \;\middle|\; \mathbf{A}\cdot\mathbf{x} \le \mathbf{b},\ \mathbf{x}\in\mathrm{RBox}(\mathbf{X})\right\}.$$

Proof: Consider the mapping $\varphi(\mathbf{z}) \triangleq \mathbf{X} - \mathbf{z}$. The mapping φ is a one-to-one and onto mapping from the set $\{\mathbf{z}\in\mathrm{RBox}(\mathbf{X}) \mid \mathbf{A}\cdot\mathbf{z} \ge \mathbf{b}\}$ to the set $\{\mathbf{x}\in\mathrm{RBox}(\mathbf{X}) \mid \mathbf{A}\cdot\mathbf{x} \le \mathbf{d}\}$. Moreover, the mapping φ satisfies $\mathbf{w}^T\cdot\mathbf{z} = \mathbf{w}^T\cdot\mathbf{X} - \mathbf{w}^T\cdot\varphi(\mathbf{z})$, and the claim follows. ■

B. Main Results

Notation: Let $\mathrm{OPT_{LP}}$ denote the set of optimal solutions of the packing LP, namely
$$\mathrm{OPT_{LP}} \triangleq \mathrm{OPT}\left(\max\left\{\mathbf{w}^T\cdot\mathbf{x} \;\middle|\; \mathbf{A}\cdot\mathbf{x}\le\mathbf{b}, \mathbf{x}\in\mathrm{RBox}(\mathbf{X})\right\}\right).$$

Algorithm 2 MIN-SUM-COVERING($\langle G, \Psi, \Phi, \mathbf{X}\rangle, t$) - a Min-Sum Algorithm for a CIP $\min\{\mathbf{w}^T \cdot \mathbf{z} \mid \mathbf{A}\cdot\mathbf{z} \ge \mathbf{b}, \mathbf{z} \in \mathrm{ZBox}(\mathbf{X})\}$. Given the Factor Graph Model $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ of the CIP and the Number of Iterations $t \in \mathbb{N}_{>0}$, Outputs a Vector $\hat{\mathbf{z}} \in \mathrm{ZBox}(\mathbf{X})$.

1) Let $\langle G, \Psi', \Phi, \mathbf{X}\rangle$ denote the factor graph model for the PLP
$$\max\left\{\mathbf{w}^T\cdot\mathbf{x} \;\middle|\; \mathbf{A}\cdot\mathbf{x}\le\mathbf{d},\ \mathbf{x}\in\mathrm{ZBox}(\mathbf{X})\right\},$$
where $\mathbf{d} \triangleq \mathbf{A}\cdot\mathbf{X} - \mathbf{b}$.
2) Let $\hat{\mathbf{x}}$ denote the outcome of MIN-SUM-PACKING($\langle G, \Psi', \Phi, \mathbf{X}\rangle, t$)
3) **return** $\hat{\mathbf{z}} \triangleq \mathbf{X} - \hat{\mathbf{x}}$.

Let $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ denote the factor graph model of this packing LP. Fix a variable vertex $r \in \mathcal{V}$ in the factor graph G of the packing LP. Let $x^{\min}_r \triangleq \min\{x^*_r \mid \mathbf{x}^* \in \mathrm{OPT_{LP}}\}$ and let $x^{\max}_r \triangleq \max\{x^*_r \mid \mathbf{x}^* \in \mathrm{OPT_{LP}}\}$. Let $\delta^{\min}_{r,t}$ and $\delta^{\max}_{r,t}$ denote the minimum and maximum values in $\delta_{r,t}$, respectively.

The proof of the following theorem appears in Section V-B.

Theorem 4 (Weak Oscillation): Consider an execution of MIN-SUM-PACKING($\langle G, \Psi, \Phi, \mathbf{X}\rangle, t$). *For every variable vertex* $r \in \mathcal{V}$ *the following holds:*

1) *If* t *is even, then* $\max\{\beta \mid \beta\in\delta_{r,t}\} \ge x^{max}_r$.
2) *If* t *is odd, then* $\min\{\beta \mid \beta\in\delta_{r,t}\} \le x^{min}_r$.

Let $\hat{x}^{(t)}_r$ denote the decision of algorithm MIN-SUM-PACKING for variable vertex r in iteration t.

Corollary 5: For every optimal solution $\mathbf{x} \in \mathrm{OPT_{LP}}$ *and every iteration* $t \in \mathbb{N}_{>0}$, *it holds that*
$$\hat{x}^{(2t+1)}_r \le x_r \le \hat{x}^{(2t)}_r.$$
Hence, if x_r *is not an integer, then* $\hat{x}^{(2t)}_r - \hat{x}^{(2t+1)}_r \ge 1$.

The following corollary implies that if algorithm MIN-SUM-PACKING outputs the same value for a vertex in two consecutive iterations, then this value is the LP optimal value.

Corollary 6: Let t *denote an even number and* s *denote an odd number. If* $\delta_{r,t}\cap\delta_{r,s} \ne \emptyset$, *then* $\delta_{r,t}\cap\delta_{r,s}$ *contains a single element* β *such that* $x^{\min}_r = x^{\max}_r = \beta$.

Proof: If $\beta \in \delta_{r,t}\cap\delta_{r,s}$, then by Theorem 4, $x^{\max}_r \le \delta^{\max}_{r,t} = \beta = \delta^{\min}_{r,s} \le x^{\min}_r$. ■

Previous works on the min-sum algorithm for optimization problems define the case that $\delta_{r,t}$ contains more than one element as a failure. Under this restricted interpretation, Theorem 4 and Corollary 5 imply a necessary condition for the convergence of the MIN-SUM-PACKING algorithm. Namely, $\mathrm{OPT_{LP}}$ must contain a unique optimal solution and this optimal solution must be integral. Indeed, if $x^{\min}_r < x^{\max}_r$, then $\hat{x}_r$ oscillates above and below the interval $(x^{\min}_r, x^{\max}_r)$ between even and odd iterations.

Analogous results holds for covering problems. We state only the theorem that is analogous to Theorem 4. Redefine $\mathrm{OPT_{LP}}$ so that it denotes the set of optimal solutions of the covering LP, i.e.,
$$\mathrm{OPT_{LP}} \triangleq \mathrm{OPT}\left(\min\left\{\mathbf{w}^T\cdot\mathbf{x} \;\middle|\; \mathbf{A}\cdot\mathbf{x}\ge\mathbf{b}, \mathbf{x}\in\mathrm{RBox}(\mathbf{X})\right\}\right).$$

Let $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ denote the factor graph model of the covering LP.

Theorem 7 (Weak Oscillation): Consider an execution of MIN-SUM-COVERING$(\langle G, \Psi, \Phi, \mathbf{X}\rangle, t)$. *For every variable vertex $r \in \mathcal{V}$ the following holds:*

1) *If t is even, then* $\min\{\beta \mid \beta \in \delta_{r,t}\} \le x_r^{min}$.
2) *If t is odd, then* $\max\{\beta \mid \beta \in \delta_{r,t}\} \ge x_r^{max}$.

See Appendix for a discussion on additional conditions needed to guarantee the convergence of the min-sum algorithm.

IV. Graph Liftings

In this section we briefly review the definition of graph coverings, state a combinatorial characterization based on [20], and show how the girth can be arbitrarily increased.

A. Covering Maps and Liftings

Definition 8 (Covering[3] Map [22]): Let $G = (V, E)$ and $\tilde{G} = (\tilde{V}, \tilde{E})$ denote finite graphs. A graph homomorphism $\pi : \tilde{G} \to G$ is a covering map *if for every $\tilde{v} \in \tilde{V}$ the restriction of π to neighbors of $\tilde{v}$ is a bijection to the neighbors of $\pi(\tilde{v})$.*

We refer only to finite covering maps. The pre-image $\pi^{-1}(v)$ of a vertex v is called the *fiber* of v. It is easy to see that all the fibers have the same cardinality if G is connected. This common cardinality is called the *degree* or *fold number* of the covering map. If $\pi : \tilde{G} \to G$ is a covering map, we call G the *base graph* and $\tilde{G}$ a *lift* of G. In the case where the fold number of the covering map is M, we say that $\tilde{G}$ is an *M-lift* of G. (see Figure 1 for illustration).

If G is connected, then every M-lift of G is isomorphic to an M-lift that is constructed as follows: (i) The vertex set is simply $\tilde{V} \triangleq V \times [M-1]$ and the covering map is the projection defined by $\pi\big((v, i)\big) \triangleq v$. (ii) For every $(u, v) \in E$, the edges in $\tilde{E}$ between the fibers of u and v constitute a matching.

The notion of M-lifts in graphs is extended to M-lifts of factor graph models in a natural manner. We denote a variable vertex in the fiber of by $\tilde{v}$ (so $\pi(\tilde{v})$ is denoted by v). Each variable vertex $\tilde{v}$ inherits the variable function of v, namely, $\tilde{w}_{\tilde{v}} \triangleq w_v$. Similarly, each constraint variable $\tilde{C}$ inherits the factor function of $\pi(\tilde{C})$. For brevity, we refer to the lifted factor graph model $\langle \tilde{G}, \tilde{\Psi}, \tilde{\Phi}, \tilde{\mathbf{X}}\rangle$ of $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ simply as the lift $\tilde{G}$ of a factor graph G.

An assignment $\mathbf{x}$ to the variable vertices $\mathcal{V}$ of a factor graph G is extended to an assignment $\tilde{\mathbf{x}}$ over the lift $\tilde{G}$ simply by defining $\tilde{x}_{\tilde{v}} \triangleq x_v$. Note that this extension preserves the validity of assignments.

Every assignment $\tilde{\mathbf{x}}$ of an M-lift $\tilde{G}$ induces an assignment of the base graph G that we call the average assignment. The *average assignment* $\mathrm{avg}(\tilde{\mathbf{x}})$ is defined by

$$\mathrm{avg}(\tilde{\mathbf{x}})_v \triangleq \frac{1}{M} \cdot \sum_{\tilde{v} \in \pi^{-1}(v)} \tilde{x}_{\tilde{v}}. \tag{6}$$

[3]The term covering is used both for optimization problems called covering problems and for topological mappings called covering maps.

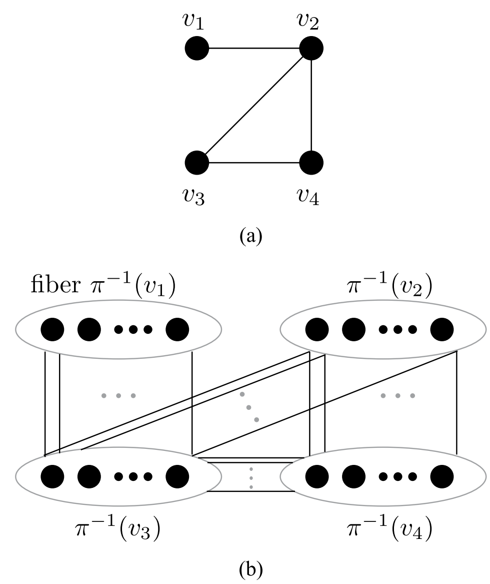

Fig. 1. An M-lift of a base graph G: (i) fiber $\pi^{-1}(v_i)$ consists of M copies of v_i, and (ii) for each edge (v_i, v_j) in G, the set of edges between fibers $\pi^{-1}(v_i)$ and $\pi^{-1}(v_j)$ is a matching. (a) Base graph G. (b) An M-lift of G.

Consider an M-lift $\tilde{G}$ of the factor graph G and a valid integral assignment $\tilde{\mathbf{x}}$ to $\tilde{G}$. By linearity, $\mathrm{avg}(\tilde{\mathbf{x}})$ is a rational valid assignment to G. The following theorem deals with the converse situation.

Theorem 9 (Special Case of [20, Th. VII.2]): For every rational feasible solution $\mathbf{x}$ of an LP, there exists an M, an M-lift $\tilde{G}$, and an integral valid assignment $\tilde{\mathbf{x}}$ to $\tilde{G}$ such that $\mathbf{x} = \mathrm{avg}(\tilde{\mathbf{x}})$.

Note that all the basic feasible solutions (extreme points) of the packing LP and the covering LP are rational.

B. Increasing Girth

The following proposition deals with obtaining lifts with large girth.

Proposition 10: There exists a finite lift $\tilde{G}$ of G such that $\mathrm{girth}(\tilde{G}) \ge 2 \cdot \mathrm{girth}(G)$.

Proof: Given a graph $G = (V, E)$, we construct a $2^{|E|}$-lift $\tilde{G} = (\tilde{V}, \tilde{E})$ as follows. Let $k = |E|$. The vertices in each fiber of $\tilde{G}$ are indexed by a binary string of length k. Index the edges in E by $\{e_1, \ldots, e_k\}$. For an edge $e_i = (u, v)$, the matching between the fiber of u and the fiber of v is induced simply by flipping the i'th bit in the index. Namely, $u_{\langle b_1 \ldots b_i \ldots b_k\rangle} \mapsto v_{\langle b_1 \ldots \overline{b_i} \ldots b_k\rangle}$.

Consider a cycle $\tilde{\gamma}$ in $\tilde{G}$ and its projection γ in G. Each edge e_i in γ must appear an even number of times. Otherwise, the i'th bit is flipped an odd number of times in $\tilde{\gamma}$, and $\tilde{\gamma}$ can not be a cycle. It follows that $\mathrm{girth}(\tilde{G}) \ge 2 \cdot \mathrm{girth}(G)$. ■

By applying Proposition 10 repeatedly, we have the following corollary for obtaining a finite lift with arbitrarily large girth.

Corollary 11: Consider a graph G. Then for any finite $\ell \in \mathbb{N}$ there exists a finite lift $\tilde{G}$ of G such that $\mathrm{girth}(\tilde{G}) \ge 2^{\ell}$.

Finally, the following corollary extends the validity of Theorem 9 to lifts with arbitrarily large girth.

Corollary 12: For every rational feasible solution $\mathbf{x}$ *of an LP and for any finite* $\ell \in \mathbb{N}$*, there exists an* M*-lift* $\tilde{G}$ *and an integral valid assignment* $\tilde{\mathbf{x}}$ *to* $\tilde{G}$ *such that (i)* $\mathbf{x} = \text{avg}(\tilde{\mathbf{x}})$*, and (ii)* $\text{girth}(\tilde{G}) > \ell$.

Proof: By Theorem 9, there exists an M'-lift $\tilde{G}'$ of G and an integral valid assignment $\tilde{\mathbf{x}}'$ to $\tilde{G}'$ such that $\mathbf{x} = \text{avg}(\tilde{\mathbf{x}}')$. Following Corollary 11, let $\tilde{G}$ denote an M''-lift of $\tilde{G}'$ such that $\text{girth}(\tilde{G}) > \ell$. Let $\tilde{\mathbf{x}}$ denote the extension of the assignment $\tilde{\mathbf{x}}'$ to variable vertices of $\tilde{G}$. That is, for every variable vertex $\tilde{v} \in \tilde{G}$, $\tilde{x}_{\tilde{v}} = \tilde{x}'_{\pi(\tilde{v})}$ where π denotes the covering map from $\tilde{G}$ to $\tilde{G}'$. Note that $\tilde{G}$ is an M-lift of G for $M = M' \cdot M''$ and that $\mathbf{x} = \text{avg}(\tilde{\mathbf{x}}') = \text{avg}(\tilde{\mathbf{x}})$ which completes the proof. ■

V. Proof of Main Results

A. Min-Sum as a Dynamic Programming on Computation Trees

Given a graph $G = (V, E)$ and a vertex $r \in V$, the path-prefix tree of height h is defined as follows.

Definition 13 (Path-Prefix Tree): Let $\hat{V}$ *denote the set of all paths of length at most* h *without backtracks that start at vertex* r*. Let*

$$\hat{E} \triangleq \big\{(p_1, p_2) \in \hat{V} \times \hat{V} \big| p_1 \text{ is a prefix of } p_2, \ |p_1| + 1 = |p_2|\big\}.$$

The directed graph $(\hat{V}, \hat{E})$ *is called the* path-prefix tree *of* G *rooted at vertex* r *with height* h*, and is denoted by* $\mathcal{T}_r^h(G)$.

We denote the zero-length path in $\hat{V}$ by (r). The graph $\mathcal{T}_r^h(G)$ is obviously acyclic and is an out-tree rooted at (r). Path-prefix trees of G that are rooted in variable vertices are often called *computation trees of* G or *unwrapped trees of* G.

We use the following notation. Vertices in a path-prefix tree $\mathcal{T}_r^h(G)$ of a graph G correspond to paths in G, and are denoted by p and q, whereas variable vertices in G are denoted by u, v, r. For a path $p \in \hat{V}$, let $t(p)$ denote the last vertex (i.e., target) of path p.

Consider a path-prefix tree $\mathcal{T}_r^h(G)$ of a factor graph $G = (\mathcal{V} \cup \mathcal{C}, E)$. We denote the vertex set of $\mathcal{T}_r^h(G)$ by $\hat{\mathcal{V}} \cup \hat{\mathcal{C}}$, where $\hat{\mathcal{V}}$ denotes paths that end in a variable vertex, and $\hat{\mathcal{C}}$ denotes paths that end in a constraint vertex. Paths in $\hat{\mathcal{V}}$ are called *variable paths*,[4] and paths in $\hat{\mathcal{C}}$ are called *constraint paths*. We attach variable functions $\hat{\phi}_p$ to variable paths p, and factor functions $\hat{\psi}_q$ to constraint paths; each vertex p inherits the function of its endpoint. The box constraint for a variable path p that ends at vertex v is defined by $\hat{X}_p \triangleq X_v$.

In the following lemma, the MIN-SUM-PACKING algorithm is interpreted as a dynamic programming algorithm over the path-prefix trees (see [16, Sec. 2]).

Lemma 14: Consider an execution of MIN-SUM-PACKING$(\langle G, \Psi, \Phi, \mathbf{X}\rangle, t)$*. Consider the computation tree* $\mathcal{T} \triangleq \mathcal{T}_r^{2t}(G) = (\hat{\mathcal{V}} \cup \hat{\mathcal{C}}, \hat{E})$*. For every variable vertex* $r \in \mathcal{V}$ *and* $\beta \in \{0, \ldots, X_r\}$*,*[5]

$$\mu_r(\beta) = \max\Bigg\{\sum_{p \in \hat{\mathcal{V}}} \hat{\phi}_p(\hat{z}_p) + \sum_{q \in \hat{\mathcal{C}}} \hat{\psi}_q(\hat{z}_{\mathcal{N}_{\mathcal{T}}(q)}) \Bigg| \begin{array}{l} \forall p \in \hat{\mathcal{V}}.\ \hat{z}_p \in \text{ZBox}(\hat{X}_p) \\ \hat{z}_{(r)} = \beta \end{array}\Bigg\}.$$

Definition 15 (Optimal Assignment): We say that a valid assignment $\hat{z}$ *to the variable paths in the computation tree* $\mathcal{T} \triangleq \mathcal{T}_r^{2t}(G) = (\hat{\mathcal{V}} \cup \hat{\mathcal{C}}, \hat{E})$ *is* optimal *if it maximizes the objective function* $\sum_{p \in \hat{\mathcal{V}}} \hat{\phi}_p(\hat{z}_p) + \sum_{q \in \hat{\mathcal{C}}} \hat{\psi}_q(\hat{z}_{\mathcal{N}_{\mathcal{T}}(q)})$.

Let $\text{OPT}_{\text{DP}}(r, t)$ denote the set of optimal valid assignments to the variable paths in $\mathcal{T}_r^{2t}(G)$. By Line 3c in algorithm MIN-SUM-PACKING, the following corollary holds.

Corollary 16: $\delta_{r,t} = \{\hat{z}_{(r)} \mid \hat{z} \in \text{OPT}_{\text{DP}}(r, t)\}$.

B. Weak Oscillation of MIN-SUM-PACKING - Proof of Theorem 4

Proof of Theorem 4: We prove the two parts of the theorem separately. We start with the proof to Part (1) of the theorem, namely, if t is even, then $\delta_{r,t}^{\max} \triangleq \max\{\beta \mid \beta \in \delta_{r,t}\} \geq \max\{x_r^* \mid \mathbf{x}^* \in \text{OPT}_{\text{LP}}\} \triangleq x_r^{\max}$.

Consider the path-prefix tree $\mathcal{T}_r^{2t}(G)$ of G rooted at variable vertex r with height $2t$. Following Corollary 16, fix an assignment $\hat{\mathbf{z}} \in \text{OPT}_{\text{DP}}(r, t)$ to variable paths in $\mathcal{T}_r^{2t}(G)$ such that $\hat{z}_{(r)} = \delta_{r,t}^{\max}$. Fix an optimal solution $\mathbf{x} \in \text{OPT}_{\text{LP}}$ such that $x_r = x_r^{\max}$ (Note that $\mathbf{x}$ is an assignment to variable vertices in G). Assume, for the sake of contradiction, that t is even and that $\hat{z}_{(r)} < x_r^{\max}$. We prove that if t is even and $\hat{z}_{(r)} < x_r^{\max}$, then there exists an assignment $\hat{\theta} \in \text{OPT}_{\text{DP}}(r, t)$ to variable paths in $\mathcal{T}_r^{2t}(G)$ such that $\hat{\theta}_{(r)} > \hat{z}_{(r)}$, a contradiction to $\hat{z}_{(r)} = \delta_{r,t}^{\max}$. Therefore, $\hat{z}_{(r)} \geq x_r^{\max}$ which completes the proof of Part (1) of the theorem.

The remainder of the proof aims towards constructing an assignment $\hat{\theta} \in \text{OPT}_{\text{DP}}(r, t)$ to variable paths in $\mathcal{T}_r^{2t}(G)$ such that $\hat{\theta}_{(r)} > \hat{z}_{(r)}$.

Without loss of generality, $\mathbf{x}$ is a basic feasible solution and hence rational. By Corollary 12, for some M, there exists an M-lift $\langle \tilde{G}, \tilde{\Psi}, \tilde{\Phi}, \tilde{\mathbf{X}}\rangle$ of $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ such that: (i) there exists an integral valid assignment $\tilde{\mathbf{x}}$ for $\tilde{G}$ such that $\mathbf{x} = \text{avg}(\tilde{\mathbf{x}})$, and (ii) $\text{girth}(\tilde{G}) > 4t$.

The value of x_r equals the average of $\tilde{\mathbf{x}}$ over the fiber of r. Let $\tilde{r}$ denote a vertex in the fiber of r such that $\tilde{x}_{\tilde{r}} \geq x_r^{\max}$.

Let $B_{\tilde{G}}(\tilde{r}, 2t)$ denote the ball of radius $2t$ centered at $\tilde{r}$. Denote by $\tilde{G}_{B(\tilde{r},2t)}$ the subgraph of $\tilde{G}$ induced by $B_{\tilde{G}}(\tilde{r}, 2t)$. Because $\text{girth}(\tilde{G}) > 4t$, $\tilde{G}_{B(\tilde{r},2t)}$ is a tree. It follows that $\tilde{G}_{B(\tilde{r},2t)}$ is isomorphic to the computation tree $\mathcal{T}_r^{2t}(G)$. Because $\hat{\mathbf{z}}$ is an optimal valid assignment to $\mathcal{T}_r^{2t}(G)$, we can regard $\hat{\mathbf{z}}$ also as an optimal valid assignment to the variable vertices in $\tilde{G}_{B(\tilde{r},2t)}$ (see Figure 2a for example).

[4]We refer by variable paths to vertices in a path-prefix tree that correspond to paths in G that end at a variable vertex.

[5]For simplicity, we re-explain notation by clarifying the expression $\hat{\psi}_q(\hat{z}_{\mathcal{N}_{\mathcal{T}}(q)})$ in the equation. Here, $\mathcal{N}_{\mathcal{T}}(q)$ equals the set of neigbors of the constraint path q in $\mathcal{T}$. Note that $\mathcal{N}_{\mathcal{T}}(q)$ contains only variable paths. The vector $\hat{z}_{\mathcal{N}_{\mathcal{T}}(q)}$ denotes the projection of the word $\hat{z}$ assigned to variable paths in $\hat{\mathcal{V}}$ onto entries associated to the variable paths in $\mathcal{N}_{\mathcal{T}}(q)$. Finally, $\hat{\psi}_q$ denotes the factor function associated with constraint path q in $\mathcal{T}$.

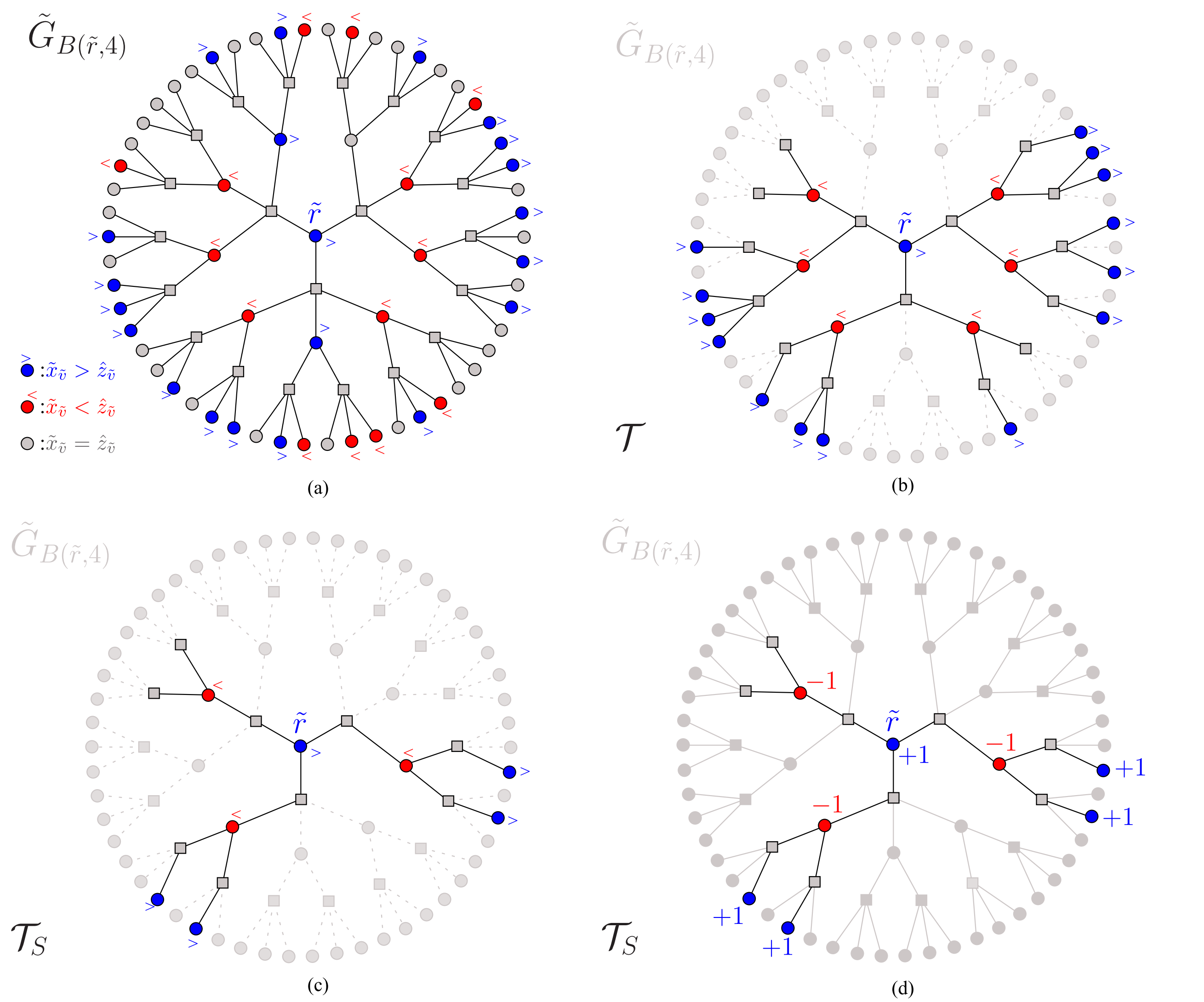

Fig. 2. Illustrations of graphs, trees and assignment in the proof of Theorem 4. Variable vertices are illustrated by circles and constraint vertices are illustrated by squares. (a) An illustration of induced subgraph $\tilde{G}_{B(\tilde{r},2t)}$ for $t = 2$. Note that $\tilde{G}_{B(\tilde{r},4)}$ is isomorphic to the computation tree $\mathcal{T}_r^4(G)$. Variable vertices $\tilde{v}$ for which $\tilde{x}_{\tilde{v}} > \hat{z}_{\tilde{v}}$ are colored in blue and marked with $>$. Variable vertices $\tilde{v}$ for which $\tilde{x}_{\tilde{v}} < \hat{z}_{\tilde{v}}$ are colored in red and marked with $<$. Variable vertices $\tilde{v}$ for which $\tilde{x}_{\tilde{v}} = \hat{z}_{\tilde{v}}$ are colored in gray. (b) The alternating tree $\mathcal{T}$ rooted at $\tilde{r}$. Paths from the root $\tilde{r}$ alternate between variable vertices $\tilde{v} \in \mathcal{E}$ for which $\tilde{x}_{\tilde{v}} > \hat{z}_{\tilde{v}}$ and variable vertices $\tilde{v} \in \mathcal{O}$ for which $\tilde{x}_{\tilde{v}} < \hat{z}_{\tilde{v}}$. (c) A skinny subtree $\mathcal{T}_S$ of the alternating tree $\mathcal{T}$. (d) Illustration of a valid assignment $\hat{\theta}$ for variable vertices in $\tilde{G}_{B(\tilde{r},2t)}$ (isomorphic to the computation tree $\mathcal{T}_r^{2t}(G)$). The values of assignment $\hat{\theta}$ are depicted with respect to the assignment $\hat{\mathbf{z}}$ and the skinny subtree $\mathcal{T}_S$, i.e., ± 1 or with no change.

By the assumption that $\hat{z}_{\tilde{r}} < x_r^{\max}$, and the choice of $\tilde{r}$ such that $x_r^{\max} \le \tilde{x}_{\tilde{r}}$, it follows that the restriction of $\tilde{\mathbf{x}}$ to the variable vertices in $\tilde{G}_{B(\tilde{r},2t)}$ does not equal $\hat{\mathbf{z}}$. In particular, it holds that $\hat{z}_{\tilde{r}} < \tilde{x}_{\tilde{r}}$. We use this difference between $\tilde{\mathbf{x}}$ and $\hat{\mathbf{z}}$ to construct two "hybrid" optimal integral solutions, one of which is the desired $\hat{\theta}$.

Let $\mathcal{E} \triangleq \{\tilde{u} \in B(\tilde{v}, 2t) \mid \frac{1}{2} \cdot d(\tilde{u}, \tilde{r}) \text{ is even}\}$. Similarly, let $\mathcal{O} \triangleq \{\tilde{u} \in B(\tilde{r}, 2t) \mid \frac{1}{2} \cdot d(\tilde{u}, \tilde{r}) \text{ is odd}\}$. Note that both $\mathcal{E}$ and $\mathcal{O}$ contain only variable vertices. We refer to $\mathcal{E}$ as the *even layers* and to $\mathcal{O}$ as the *odd layers*.

Let $\mathcal{F}$ denote the subgraph of $\tilde{G}_{B(\tilde{r},2t)}$ that is induced by: (i) the vertices $\tilde{u} \in \mathcal{E}$ such that $\hat{z}_{\tilde{u}} < \tilde{x}_{\tilde{u}}$, (ii) the vertices $\tilde{u} \in \mathcal{O}$ such that $\hat{z}_{\tilde{u}} > \tilde{x}_{\tilde{u}}$, and (iii) constraint vertices in $\tilde{G}_{B(\tilde{r},2t)}$. By definition, $\tilde{r}$ is a vertex in $\mathcal{E}$, because by our assumption $\hat{z}_{\tilde{r}} < \tilde{x}_{\tilde{r}}$. Hence, $\tilde{r} \in \mathcal{F}$. Let $\mathcal{T}$ denote the connected component of $\mathcal{F}$ that contains $\tilde{r}$. We root $\mathcal{T}$ at $\tilde{r}$, and refer to $\mathcal{T}$ as an *alternating tree* (see Figure 2b).

A subtree $\mathcal{T}_S$ of $\mathcal{T}$ is a *skinny tree* if each constraint vertex chooses at most one child and each variable vertex chooses all its children (see Figure 2c). Formally, a subtree $\mathcal{T}_S$ of $\mathcal{T}$ is a skinny tree if it is a maximal tree with respect to inclusion among all trees that satisfy (i) $\tilde{r} \in \mathcal{T}_S$, (ii) $\deg_{\mathcal{T}_S}(\tilde{C}) = 2$ for every constraint vertex $\tilde{C}$ in $\mathcal{T}_S$, and (iii) $\deg_{\mathcal{T}_S}(\tilde{u}) = \deg_{\mathcal{T}}(\tilde{u})$ for every variable vertex $\tilde{u}$ in $\mathcal{T}_S$. We fix a skinny subtree $\mathcal{T}_S$ of $\mathcal{T}$. The skinny subtree $\mathcal{T}_S$ serves as a pivot for obtaining two "hybrid" optimal integral solutions, one of which is $\hat{\theta}$ that results to the contradiction argument.

For a subset of variable vertices $\tilde{A} \subseteq \tilde{\mathcal{V}}$, let $\tilde{w}(\tilde{A}) \triangleq \sum_{\tilde{u} \in \tilde{A}} w_u$ (recall that the weight w_u of a variable vertex u in G is given to each vertex $\tilde{u}$ in the fiber of u). We rely on the following claim (which we prove later).

Claim 17:

$$\tilde{w}\big(\mathcal{T}_S \cap \mathcal{E}\big) \geq \tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big). \tag{7}$$

We now define an assignment $\hat{\theta}$ to variable vertices in $\tilde{G}_{B(\tilde{r},2t)}$ by

$$\hat{\theta}_{\tilde{u}} \triangleq \begin{cases} \hat{z}_{\tilde{u}} + 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{E} \\ \hat{z}_{\tilde{u}} - 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{O} \\ \hat{z}_{\tilde{u}} & \text{otherwise.} \end{cases}$$

We claim that $\hat{\theta}$ is a valid integral assignment for $\tilde{G}_{B(\tilde{r},2t)}$ (see Figure 2d). The proof is analogous to the proof that $\tilde{\mathbf{y}}$ is a valid assignment in the proof of Claim 17. By Equation (7), the value of $\hat{\theta}$ is not less than the value of $\hat{\mathbf{z}}$ since

$$\sum_{\tilde{u} \in \tilde{\mathcal{V}} \cap B_{\tilde{G}}(\tilde{r},2t)} \big(\phi_{\tilde{u}}(\hat{\theta}_{\tilde{u}}) - \phi_{\tilde{u}}(\hat{z}_{\tilde{u}})\big) = \tilde{w}\big(\mathcal{T}_S \cap \mathcal{E}\big) - \tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big) \geq 0. \tag{8}$$

Therefore, $\hat{\theta} \in \mathrm{OPT_{DP}}(r,t)$. However, $\hat{\theta}_{\tilde{r}} > \hat{z}_{\tilde{r}} = \delta^{\max}_{r,t}$, a contradiction. It follows that $\delta^{\max}_{r,t} \geq x^{\max}_{r,t}$ if t is even.

The proof of Part (2) of the theorem that $\delta^{\min}_{r,t} \leq x^{\min}_{r}$ for an odd t is analogous to the proof that $\delta^{\max}_{r,t} \geq x^{\max}_{r}$ if t is even. It requires the following modifications:

a. Fix $\hat{\mathbf{z}} \in \mathrm{OPT_{DP}}(r,t)$ such that $\hat{z}_{(r)} = \delta^{\min}_{r,t}$ and $\mathbf{x} \in \mathrm{OPT_{LP}}$ such that $x_r = x^{\min}_r$.
b. Assume towards a contradiction that t is odd and that $\hat{z}_{(r)} > x_r$.
c. Pick $\tilde{r}$ so that $\tilde{x}_{\tilde{r}} \leq x^{\min}_r$.
d. The forest $\mathcal{F}$ is induced by the following set of vertices: (i) the vertices $\tilde{u} \in \mathcal{E}$ such that $\hat{z}_{\tilde{u}} > \tilde{x}_{\tilde{u}}$, (ii) the vertices $\tilde{u} \in \mathcal{O}$ such that $\hat{z}_{\tilde{u}} < \tilde{x}_{\tilde{u}}$, and (iii) constraint vertices in $\tilde{G}_{B(\tilde{r},2t)}$.
e. Prove that the weight of even layers in the skinny tree is not greater than the weight of the odd layers.
f. The assignment $\tilde{\mathbf{y}}$ is defined so that it increments even layers and decrements odd layers.
g. The assignment $\hat{\theta}$ is defined so that it decrements even layers and increments odd layers.

■

Proof of Claim 17: To prove Equation (7), define an integral assignment $\tilde{\mathbf{y}}$ to variable vertices in $\tilde{\mathcal{V}}$ by

$$\tilde{y}_{\tilde{u}} \triangleq \begin{cases} \tilde{x}_{\tilde{u}} - 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{E} \\ \tilde{x}_{\tilde{u}} + 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{O} \\ \tilde{x}_{\tilde{u}} & \text{otherwise.} \end{cases}$$

Observe that $\tilde{\mathbf{y}}$ is a valid assignment for $\tilde{G}$. Indeed, all the box constraints are satisfied because we increment a value compared to $\tilde{x}_u$ only if $\tilde{x}_{\tilde{u}} < \hat{z}_{\tilde{u}}$. Similarly, we decrement a value compared to $\tilde{x}_u$ only if $\tilde{x}_{\tilde{u}} > \hat{z}_{\tilde{u}}$. In addition, we need to show that every constraint is satisfied by $\tilde{\mathbf{y}}$. Note that a constraint $\tilde{C}$ may have at most two neighbors that are variable vertices in the skinny tree; the rest of the neighbors retain the value assigned by $\tilde{\mathbf{x}}$. If a constraint $\tilde{C}$ is not a neighbor of a variable vertex in $\mathcal{T}_S$, then it is satisfied because $\tilde{\mathbf{x}}$ satisfies it. If a constraint $\tilde{C}$ has two neighbors that are variable vertices in $\mathcal{T}_S$, then one is incremented and the other is decremented. Overall, the constraint remains satisfied. Finally, suppose $\tilde{C}$ has only one neighbor in $\mathcal{T}_S$. Denote this neighbor by $\tilde{v}$. Then $\tilde{v}$ is a parent of $\tilde{C}$. If $\tilde{y}_{\tilde{v}} = \tilde{x}_{\tilde{u}} - 1$, then clearly $\tilde{C}$ is satisfied by $\tilde{\mathbf{y}}$. If the value assigned to $\tilde{v}$ is incremented, then $\tilde{v} \in \mathcal{O}$ (i.e., an odd layer). This implies that the children of $\tilde{C}$ are in an even layer, the distance of which to the root is at most $2t$. Hence, the children of $\tilde{C}$ belong to the ball $B(\tilde{r},2t)$. Moreover, these children do not belong to the alternating tree (otherwise, one of its children would belong to the skinny tree). Thus, for each child $\tilde{u}$ of $\tilde{C}$ we have $\hat{z}_{\tilde{u}} \geq \tilde{x}_{\tilde{u}} = \tilde{y}_{\tilde{u}}$. In addition, $\hat{z}_{\tilde{v}} \geq \tilde{y}_{\tilde{v}}$. Hence, $\hat{\mathbf{z}} \geq \tilde{\mathbf{y}}$ when restricted to the neighbors of $\tilde{C}$. Because $\hat{\mathbf{z}}$ satisfies $\tilde{C}$, so does $\tilde{\mathbf{y}}$, as required. Because $\tilde{\mathbf{y}}$ is a valid assignment, $\mathrm{avg}(\tilde{\mathbf{y}})$ is a feasible solution of the packing LP. The optimality of $\mathbf{x}$ implies that $\mathbf{w}^T \cdot \mathbf{x} \geq \mathbf{w}^T \cdot \mathrm{avg}(\tilde{\mathbf{y}})$. By the definition of $\tilde{\mathbf{y}}$, we have

$$\mathbf{w}^T \cdot \big(\mathbf{x} - \mathrm{avg}(\tilde{\mathbf{y}})\big) = \frac{1}{M} \cdot \big(\tilde{w}(\mathcal{T}_S \cap \mathcal{E}) - \tilde{w}(\mathcal{T}_S \cap \mathcal{O})\big),$$

and Equation (7) follows. ■

APPENDIX
ON CONVERGENCE OF THE MIN-SUM ALGORITHM FOR NONBINARY PACKING AND COVERING PROBLEMS

In Section III-B we showed that if the MIN-SUM-PACKING algorithm outputs the same result in two consecutive iterations, then this result equals the optimal solution of the LP relaxation (see Corollary 6). On the other hand, even if the LP relaxation has a unique optimal solution and that solution is integral, then the MIN-SUM-PACKING algorithm may not converge (see Sanghavi *et al.* [14] for an example with respect to the maximum weight independent set problem).

Convergence of the min-sum algorithm was proved for the maximum weight b-matching and the minimum r-edge covering problems by Sanghavi *et al.* [13] and Bayati *et al.* [12]. They considered the zero-one integer program for maximum weight matching with the constraints $\sum_{e \ni v} x_e \leq 1$, and proved that after a pseudo-polynomial number of iterations, the min-sum algorithm converges to the optimal solution of the LP relaxation provided that it is unique and integral. The parameter that is used to bound the number of iterations is defined as follows.

Definition 18 [13]: Given a polyhedron $\mathcal{P} \subseteq \mathbb{R}^n$ and a cost vector $\mathbf{w} \in \mathbb{R}^n$. Define $c(\mathcal{P}, \mathbf{w})$ by

$$c(\mathcal{P}, \mathbf{w}) \triangleq \min_{\mathbf{x} \in \mathcal{P} \setminus \{\mathbf{x}^*\}} \frac{\mathbf{w}^T \cdot (\mathbf{x}^* - \mathbf{x})}{\|\mathbf{x}^* - \mathbf{x}\|_1},$$

where $\mathbf{x}^* \in \mathrm{OPT}\big(\max\{\mathbf{w}^T \cdot \mathbf{x} \mid \mathbf{x} \in \mathcal{P}\}\big)$.

By definition, $C(\mathcal{P}, \mathbf{w}) \geq 0$, and $c(\mathcal{P}, \mathbf{w}) > 0$ if and only the LP has a unique optimal solution. On the other hand, $c(\mathcal{P}, \mathbf{w}) \leq w_{\max}$, where $w_{\max} \triangleq \max\{w_i \mid 1 \leq i \leq n\}$.

We generalize the convergence result of Sanghavi *et al.* [13] to nonbinary packing and covering problems. For the sake of brevity we state the result for packing problems; the analogous result for covering problems holds as well.

Theorem 19: Let $\mathcal{P}$ denote the polytope $\{\mathbf{x} \in \mathrm{RBox}(\mathbf{X}) \mid \mathbf{A} \cdot \mathbf{x} \leq \mathbf{b}\}$. Assume that every column of $\mathbf{A}$ contains at most *two* 1*s. Assume that the packing LP* $\max\{\mathbf{w}^T \cdot \mathbf{x} \mid \mathbf{x} \in \mathcal{P}\}$ *has a unique optimal solution* $\mathbf{x}^*$ *such that* $x_i^* \in \{0, X_i\}$ *for every* $1 \leqslant i \leqslant n$*. Let* $\langle G, \Psi, \Phi, \mathbf{X}\rangle$ *denote the factor graph model of the packing LP. If* $t > \frac{w_{\max}}{c(\mathcal{P},w)} + \frac{1}{2}$*, then the output* $\hat{\mathbf{x}}$ *of Algorithm* MIN-SUM-PACKING$(\langle G, \Psi, \Phi\rangle, t)$ *satisfies* $\hat{\mathbf{x}} = \mathbf{x}^*$.

We first explain why the two restricting assumptions in Theorem 19 are useful.

Observation 20: If each column of $\mathbf{A}$ *contains at most two* 1*s, then the degrees of the variable vertices in the factor graph are at most* 2*. Hence, the alternating skinny tree (as in the proof of Theorem 4) reduces to a path.*

For every M-lift $\tilde{G}$ of a factor graph G, we define the polytope $\tilde{\mathcal{P}} = \{\tilde{\mathbf{x}} \in \mathrm{RBox}(\tilde{\mathbf{X}}) \mid \tilde{\mathbf{A}} \cdot \tilde{\mathbf{x}} \leq \tilde{\mathbf{b}}\}$, where $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{b}}$ are the constraint matrix and vector of the lifted factor graph.

Observation 21: If the unique solution $\mathbf{x}^*$ *of the packing LP satisfies* $x_i^* \in \{0, X_i\}$ *for every* $1 \leqslant i \leqslant n$*, then* $C(\tilde{\mathcal{P}}, \tilde{\mathbf{w}}) = C(\mathcal{P}, \mathbf{w})$ *for every* M*-lift* $\tilde{G}$ *of the factor graph* G*.*

Proof of Theorem 19: We focus on the case that t is even; the proof for odd values of t is analogous. The proof uses many of the notions used in the proof of Theorem 4 with modifications based on Observations 20 and 21, hence we reuse the same notations. Note that if t is even and $x_r^* = X_r$, then Theorem 4 implies that $\delta_{r,t}^{\max} = X_r$, and hence the output of algorithm MIN-SUM-PACKING satisfies $\hat{x}_r = x_r^*$, as required. Thus, we are left with the case that $x_r^* = 0$ and wish to prove that $\delta_{r,t}^{\max} = 0$ if t is even.

Assume towards a contradiction that $\delta_{r,t}^{\max} > 0$, and let $\hat{\mathbf{z}} \in \mathrm{OPT}_{\mathrm{DP}}(r,t)$ denote an assignment such that $\hat{z}_{(r)} = \delta_{r,t}^{\max} > 0$. As in the proof of Theorem 4, we define an alternating tree in $\tilde{G}_{B(\tilde{r},2t)}$. However, the variable vertices in the even layers of the alternating tree satisfy $\hat{z}_{\tilde{v}} > \tilde{x}_{\tilde{v}}^*$. Variables vertices in the odd layers of the alternating tree satisfy $\hat{z}_{\tilde{v}} < \tilde{x}_{\tilde{v}}^*$. By Observation 20, the skinny tree is simply a path that contains the root $\tilde{r}$.

Let L denote the set of leaves of the skinny tree whose distance from $\tilde{r}$ is $2t$. Because the skinny tree is a path, it follows that $|L| \leq 2$.

Define an integral assignment $\tilde{\mathbf{y}}$ to variable vertices in $\tilde{\mathcal{V}}$ by

$$\tilde{y}_{\tilde{u}} \triangleq \begin{cases} \tilde{x}_{\tilde{u}}^* + 1 & \text{if } \tilde{u} \in (\mathcal{T}_S \cap \mathcal{E}) \setminus L \\ \tilde{x}_{\tilde{u}}^* - 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{O} \\ \tilde{x}_{\tilde{u}}^* & \text{otherwise.} \end{cases}$$

As in the proof of Claim 17, the assignment $\tilde{\mathbf{y}}$ is a valid assignment for $\tilde{G}$.

Assume that $L = \emptyset$. Since $\tilde{x}^*$ is a unique optimal assignment, it follows that

$$\tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big) > \tilde{w}\big((\mathcal{T}_S \cap \mathcal{E}) \setminus L\big). \tag{9}$$

Define

$$\hat{\theta}_{\tilde{u}} \triangleq \begin{cases} \hat{z}_{\tilde{u}} - 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{E} \\ \hat{z}_{\tilde{u}} + 1 & \text{if } \tilde{u} \in \mathcal{T}_S \cap \mathcal{O} \\ \hat{z}_{\tilde{u}} & \text{otherwise.} \end{cases} \tag{10}$$

As in the proof of Theorem 4, the assignment $\hat{\theta}$ is a valid integral assignment for $\tilde{G}_{B(\tilde{r},2t)}$. But, Equation (9) implies that the assignment $\hat{\theta}$ has a higher value than $\hat{\mathbf{z}}$. This contradicts the optimality of $\hat{\mathbf{z}}$.

Assume that $|L| = 2$ (the proof for $|L| = 1$ is similar). By the definition of $C(\tilde{\mathcal{P}}, \tilde{\mathbf{w}})$ and by Observation 21,

$$\frac{\tilde{\mathbf{w}}^T \cdot (\tilde{\mathbf{x}}^* - \tilde{\mathbf{y}})}{\|\tilde{\mathbf{x}}^* - \tilde{\mathbf{y}}\|_1} \geq C(\mathcal{P}, \mathbf{w}). \tag{11}$$

Note that

$$\tilde{\mathbf{w}}^T \cdot (\tilde{\mathbf{x}}^* - \tilde{\mathbf{y}}) = \tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big) - \tilde{w}\big((\mathcal{T}_S \cap \mathcal{E}) \setminus L\big),$$

and

$$\|\tilde{\mathbf{x}}^* - \tilde{\mathbf{y}}\|_1 = |\mathcal{T}_S| - |L| = 2t - 1.$$

It follows that

$$\tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big) - \tilde{w}\big((\mathcal{T}_S \cap \mathcal{E}) \setminus L\big) \geq (2t-1) \cdot C(\mathcal{P}, \mathbf{w}). \tag{12}$$

Consider the assignment $\hat{\theta}$ defined in Equation (10). By Equation (12) we have

$$\begin{aligned} &\sum_{\tilde{u} \in \tilde{\mathcal{V}} \cap B_{\tilde{G}}(\tilde{r},2t)} \big(\phi_{\tilde{u}}(\hat{\theta}_{\tilde{u}}) - \phi_{\tilde{u}}(\hat{z}_{\tilde{u}})\big) \\ &\quad = \tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big) - \tilde{w}\big(\mathcal{T}_S \cap \mathcal{E}\big) \\ &\quad = \tilde{w}\big(\mathcal{T}_S \cap \mathcal{O}\big) - \tilde{w}\big((\mathcal{T}_S \cap \mathcal{E}) \setminus L\big) - \tilde{w}\big(L\big) \\ &\quad \geq (2t-1) \cdot C(\mathcal{P}, \mathbf{w}) - 2 \cdot w_{\max}. \end{aligned}$$

Because $t > \frac{w_{\max}}{c(\mathcal{P},w)} + \frac{1}{2}$, it follows that the assignment $\hat{\theta}$ has a higher value than $\hat{\mathbf{z}}$. This contradicts the optimality of $\hat{\mathbf{z}}$. It follows that $\delta_{r,t}^{\max} = 0$ if t is even. ■

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